



2020 YEAR 11 ASSESSMENT TASK 3

Mathematics Advanced

Thursday 10th September

General**Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
100**Section I – 10 marks**

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks

- Attempt Questions 11– 40
- Allow about 2 hours and 45 minutes for this section

Name: _____

Student Number: _____

Outcomes

A student:

- | | |
|--------|---|
| MA11-1 | uses algebraic and graphical techniques to solve, and where appropriate, compare alternative solutions to problems |
| MA11-2 | uses the concepts of functions and relations to model, analyse and solve practical problems |
| MA11-3 | uses the concepts and techniques of trigonometry in the solution of equations and problems involving geometric shapes |
| MA11-4 | uses the concepts and techniques of periodic functions in the solutions of trigonometric equations or proof of trigonometric identities |
| MA11-5 | interprets the meaning of the derivative, determines the derivative of functions and applies these to solve simple practical problems |
| MA11-6 | manipulates and solves expressions using the logarithmic and index laws, and uses logarithms and exponential functions to solve practical problems |
| MA11-7 | uses concepts and techniques from probability to present and interpret data and solve problems in a variety of contexts, including the use of probability distributions |
| MA11-9 | provides reasoning to support conclusions which are appropriate to the context |

Section I

10 marks

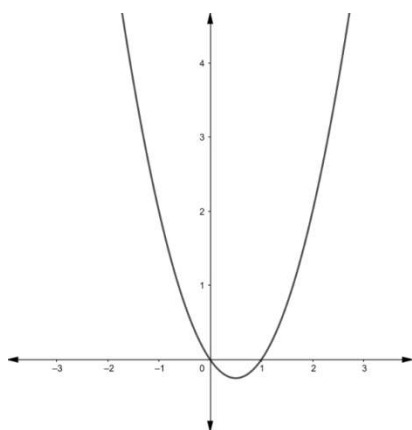
Attempt Questions 1–10

Allow about 15 minutes for this section

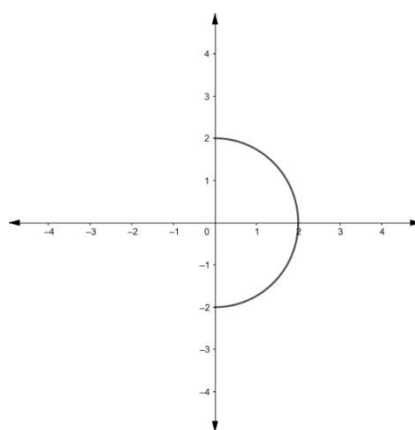
Use the multiple-choice answer sheet for Questions 1–10

1 Which graph represents a one-to-one function?

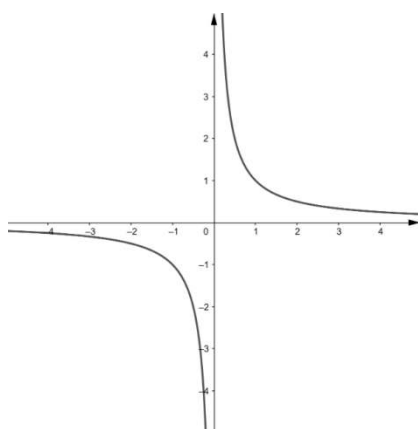
A.



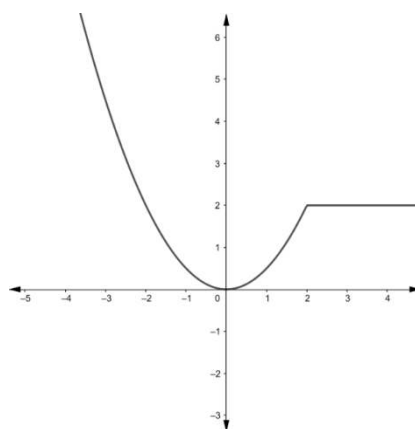
B.



C.



D.



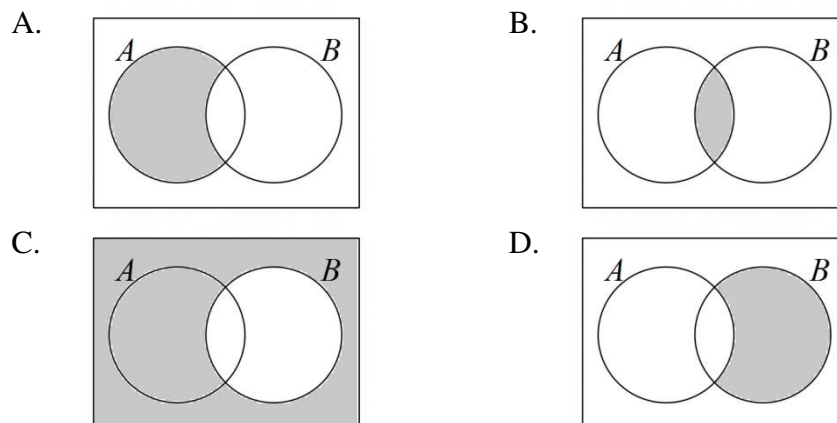
2 If $f(x) = x^2 - 2x$, what is the value of $f(-3)$?

- A. -15
- B. -3
- C. 0
- D. 15

3 How many x -intercepts does the graph of $f(x) = (x-2)(x^2 + 4x + 3)$ have?

- A. 0
- B. 1
- C. 2
- D. 3

4 In which of the Venn diagrams is $A \cap B'$ shaded?



5 Which of the following lines is parallel to the line $6x - 2y - 3 = 0$?

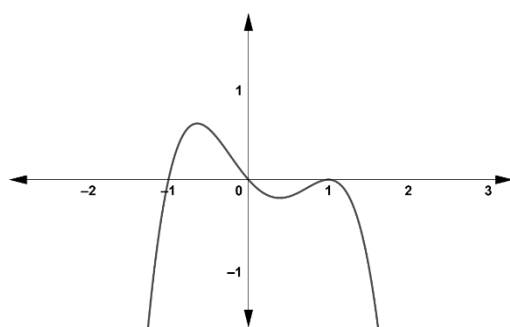
- A. $6x - y - 5 = 0$
- B. $y = 3x - 5$
- C. $y = \frac{1}{3}x - 8$
- D. $y = -3x + 5$

6 What are the solutions to $|2x + 3| = 1$?

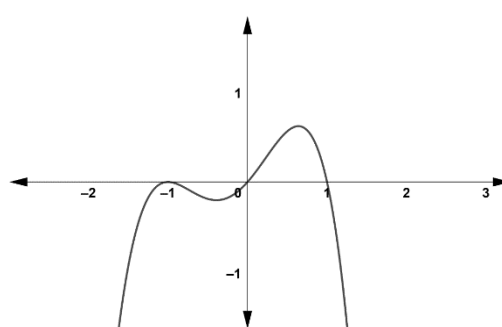
- A. $x = -2$ or $x = -1$
- B. $x = -2$ or $x = 1$
- C. $x = 1$ or $x = 2$
- D. $x = 2$ or $x = -1$

7 Which of the following graphs shows the function $y = -x(x-1)^2(x+1)$?

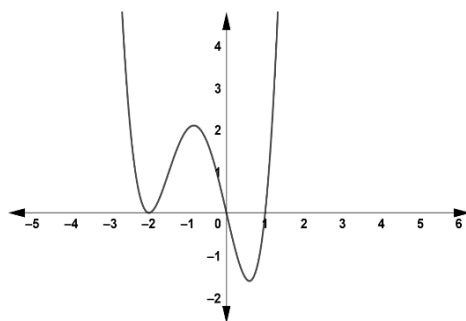
A.



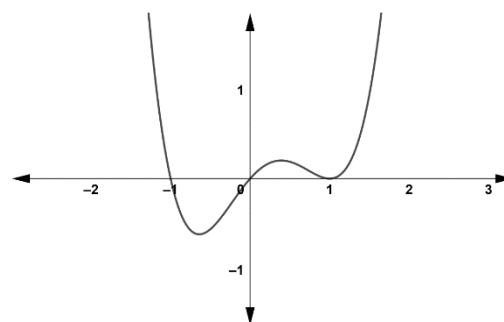
B.



C.



D.



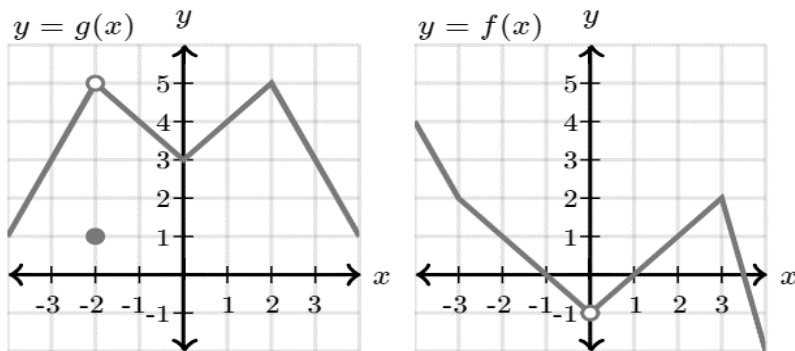
8 Which of the following is always true for $0^\circ < \theta < 90^\circ$?

- A. $\sin(90 - \theta) = \sin \theta$
- B. $\sin(180 - \theta) = \sin \theta$
- C. $\sin(180 + \theta) = \sin \theta$
- D. $\sin(360 - \theta) = \sin \theta$

9 What are the solutions to $\sin x = -\frac{1}{2}$ for $0 \leq x \leq 360^\circ$?

- A. $x = 30^\circ$
- B. $x = 30^\circ$ and $x = 150^\circ$
- C. $x = 150^\circ$ and $x = 210^\circ$
- D. $x = 210^\circ$ and $x = 330^\circ$

10 The functions $f(x)$ and $g(x)$ are graphed below.
What is the value of $g(f(x))$ when $x = -2$?



- A. 0
- B. 1
- C. 4
- D. The value does not exist.

Student Number: _____

2020 YEAR 11 ASSESSMENT TASK 3

Mathematics Advanced

Section II Answer Booklet

90 marks

Attempt Questions 11 - 40

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing paper is provided at the back of each booklet. If you use this space, clearly indicate which question you are answering.
-

Question 11 (2 marks)

Simplify $\frac{2x^2 - 5x - 12}{x - 4}$.

2

.....

.....

.....

.....

.....

.....

.....

.....

Question 12 (3 marks)

Write $\frac{20}{x^2 + 5x} - \frac{4}{x}$ as a single fraction in simplest form.

3

.....

.....

.....

.....

.....

.....

.....

.....

Question 13 (1 mark)

Evaluate $\log_a a^2$.

1

.....

.....

Question 14 (2 marks)

Solve $\log_2(x+2) = 3$.

2

.....

.....

.....

.....

.....

.....

Question 15 (1 mark)

Convert 225° to radians in exact form.

1

.....

.....

.....

.....

Question 16 (2 marks)

State the natural domain of $f(x) = \frac{5}{\sqrt{5-x}}$ using interval notation.

2

.....

.....

.....

.....

.....

.....

Question 17 (2 marks)

Determine whether the function $f(x) = x^3 + 2x$ is odd, even or neither and justify your answer. **2**

[illegible]

Question 18 (2 marks)

A bag contains 4 yellow and 3 red balls. Two balls are chosen at random and the first ball is not replaced before the second ball is chosen. Find the probability that one ball of each colour is chosen.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 19 (2 marks)

Given $\sin \theta = \frac{5}{6}$ and $90^\circ \leq \theta \leq 180^\circ$, find the exact value of $\cos \theta$.

2

.....

.....

.....

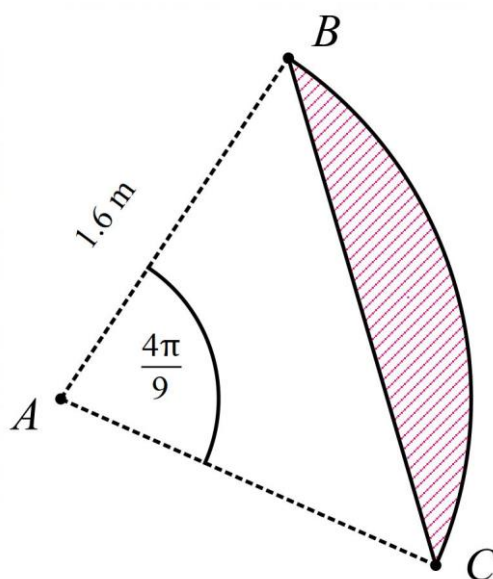
.....

.....

Question 20 (2 marks)

The shaded segment shown is formed by an arc BC of a circle with radius 1.6 metres and the chord BC which subtends an angle of $\frac{4\pi}{9}$ radians at its centre A . Find the area of the shaded segment, correct to 3 significant figures.

2



.....

.....

.....

.....

.....

.....

Questions 11 – 20 are worth 19 marks in total

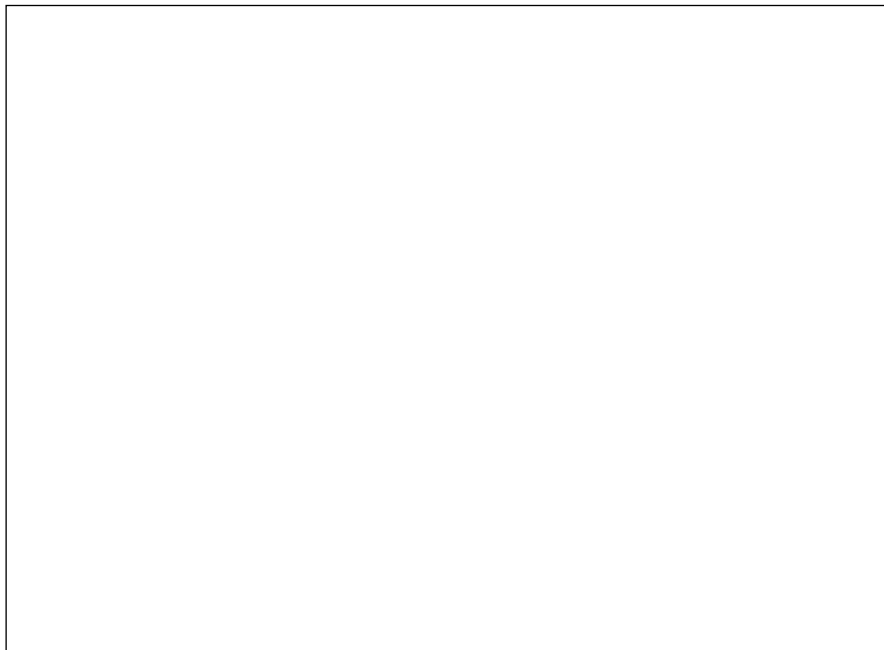
Student Number: _____

Question 21 (4 marks)

- (a) Sketch the graph of the function $f(x) = x^2 - 5x + 6$ showing all intercepts and important features. **2**



- (b) Sketch the graph of the function $f(x) = 1 + 3^x$ showing all important features. **2**



Question 22 (4 marks)

The table shows the value of tips Isabella receives when she works in a restaurant on a Friday night. The probabilities of each value being received are based on experience.

Tips (to the nearest Dollar)	\$5	\$12	\$23	\$35	\$38	\$40
Probability	0.15	0.26	0.3	0.16	0.1	0.03

- (a) Explain why the above table is a probability distribution. **2**

.....

.....

.....

.....

- (b) Calculate the expected value of Isabella's tips for the next Friday she works. **1**

.....

.....

.....

.....

- (c) What is the percentage probability that Isabella will receive at least \$35 worth of tips? **1**

.....

.....

.....

.....

Question 23 (2 marks)

Sketch the graph of $y = (x - 4)^2(x + 1)$.

2



Question 24 (4 marks)

- a) Find the centre and the radius of the circle represented by the equation $x^2 - 6x + y^2 + 2y + 1 = 0$. **2**

.....

.....

.....

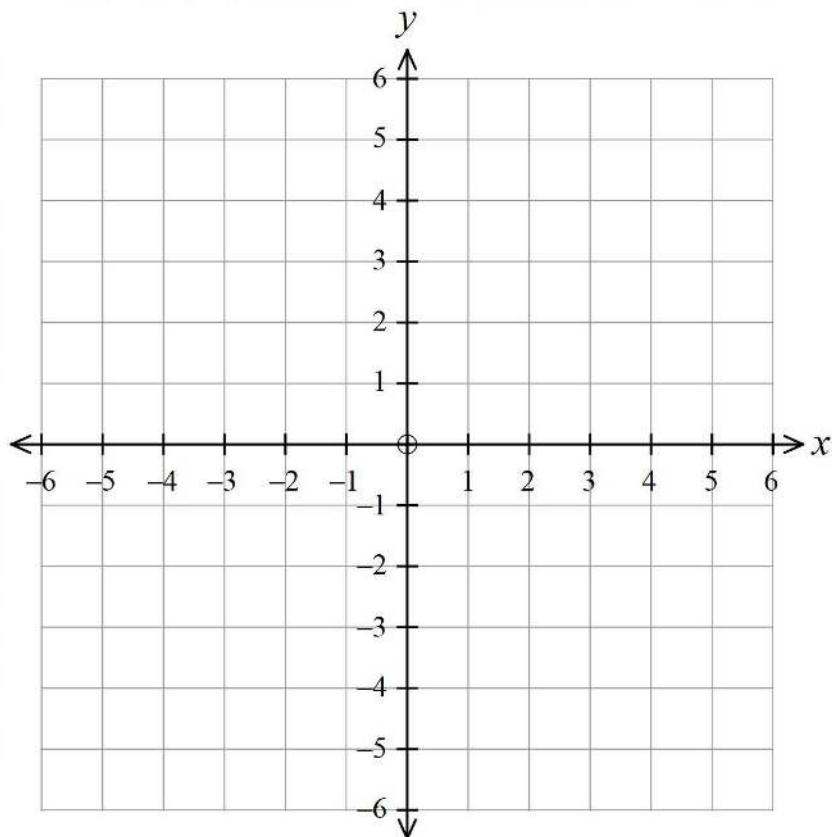
.....

.....

.....

.....

- (b) Draw the circle on the number plane below. You do **not** need to find any x -intercepts. **2**



Question 25 (5 marks)

- (a) Find the equation of the tangent to the curve $y = 3x^3 - 10x^2 + 1$ at the point where $x = 2$. **3**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Find the angle of inclination between the tangent and the positive x -axis to the nearest degree. **2**

.....

.....

.....

.....

.....

.....

.....

.....

Questions 11 – 25 are worth 38 marks in total
--

Student Number: _____

Question 26 (2 marks)

At a Performing Arts College 70% of students study Drama, 55% of students study Music and 35% of students study both Drama and Music.

- (a) What is the probability that a randomly selected student studies Music or Drama or both? 1

.....
.....
.....

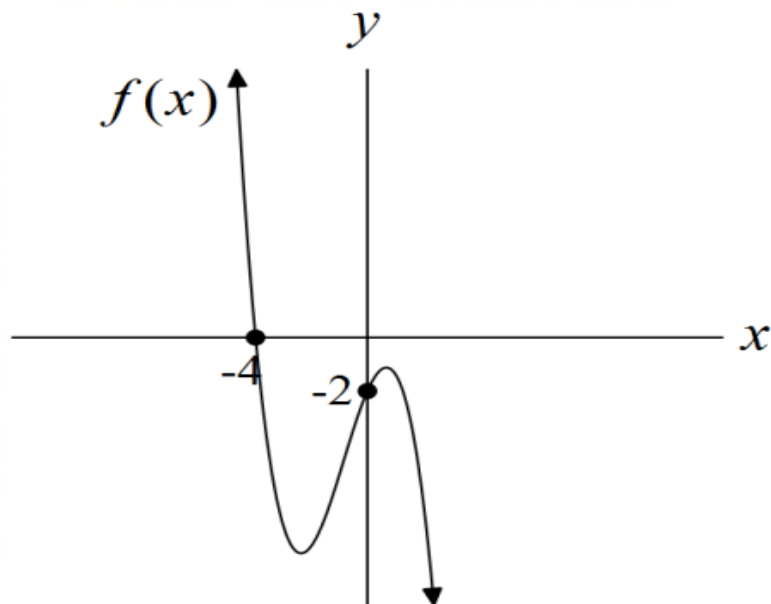
- (b) Given that a student studies Drama what is the probability that this student also studies Music? 1

.....
.....
.....
.....

Question 27 (2 marks)

The diagram shows the curve $y = f(x)$.

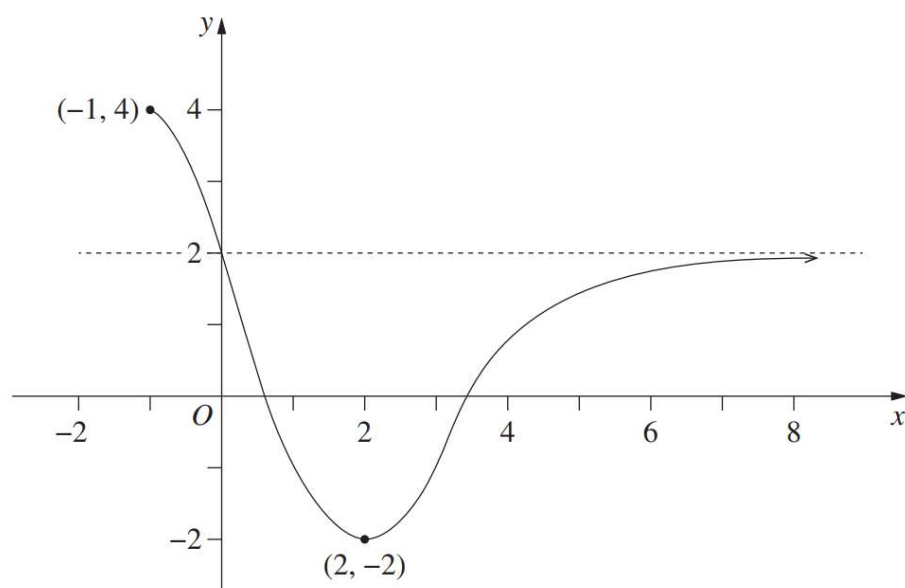
Sketch $y = f(-x)$ on the same set of axes, showing any x - and y -intercepts. 2



Question 28 (2 marks)

The graph of a function $y = f(x)$ is shown. It has an asymptote at $y = 2$.

2



Using interval notation, state the domain and range of $y = f(x)$.

.....

.....

.....

.....

.....

Question 29 (8 marks)

Differentiate the following with respect to x .

(a) $y = 2x(x + 7)$ **1**

.....
.....
.....

(b) $y = 3(x^2 - 1)^3$ **2**

.....
.....
.....

(c) $y = 2x\sqrt{x}$ **2**

.....
.....
.....
.....
.....

(d) $y = \frac{2x^3}{x^2 - 5}$ **3**

.....
.....
.....
.....
.....
.....
.....
.....
.....

Question 30 (2 marks)

A piecewise function $f(x)$ is defined as follows.

2

$$f(x) = \begin{cases} -x + a, & x < 1 \\ x^2 - ax + 2, & x \geq 1 \end{cases}$$

What value of a makes this function continuous?

.....

.....

.....

.....

.....

.....

.....

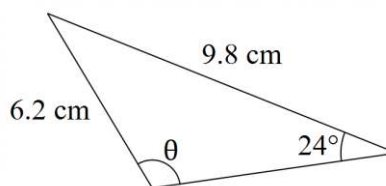
.....

.....

Question 31 (3 marks)

The diagram shows an obtuse triangle with side lengths 6.2 cm and 9.8 cm, and angles 24° and θ , where θ is the largest angle.

3



Find the value of θ to the nearest degree.

.....

.....

.....

.....

.....

.....

.....

.....

.....

Student Number: _____

Question 32 (3 marks)

The function $f(x) = 2x(x-4)^3$ has two horizontal tangents. Given that one horizontal tangent touches the curve at $(4, 0)$, find the coordinates of the other point on the curve at which the tangent is horizontal.

3

[illegible]

Question 33 (3 marks)

3

Solve $2\cos^2 x - \cos x = 0$ for $0 \leq x \leq 2\pi$.

.....

.....

.....

.....

.....

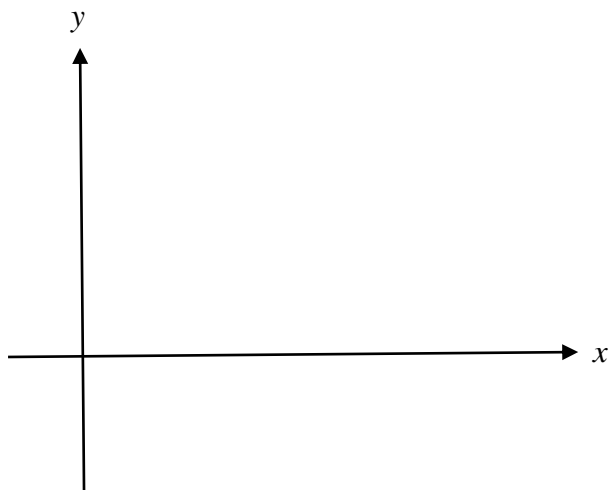
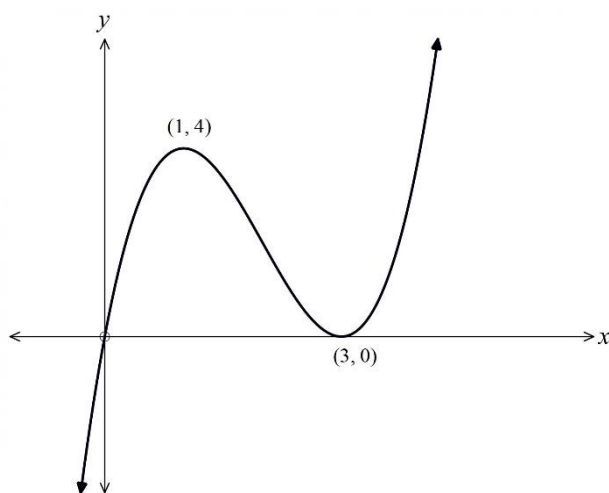
.....

.....

Question 34 (2 marks)

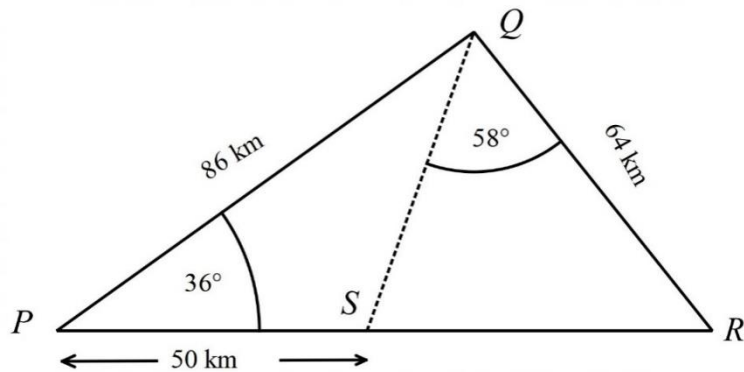
2

The diagram shows the graph of the curve $y = f(x)$. Sketch the graph of the gradient function on the number plane below.



Question 35 (4 marks)

Triangles PQS and QRS have a common side QS , as shown.



- (a) Show that the length $QS = 54.2$ km correct to 3 significant figures.

2

[illegible]

- (b) Hence or otherwise find the area of ΔPQR correct to the nearest square kilometre.

2

[illegible]

Question 36 (2 marks)

Use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ given $f(x) = x^2 - 2x$.

2

[illegible]

Question 37 (4 marks)

The decibel, dB, is a logarithmic unit used to measure sound level. The formula to find the decibel level of a sound is $\text{dB} = 10 \log_{10} \left(\frac{I}{I_o} \right)$ where I is the intensity of the sound, and I_o is the intensity of the threshold of sound (sound which can barely be perceived).

- (a) According to Safe Work Australia, workers should wear ear protection if they are exposed to sound levels of 85 dB or more for an extended period of time. A machine produces sound at an intensity of $(1.8 \times 10^{10})I_o$. **2**

Should workers who spend time near this machine wear ear protection?
Justify your answer with calculations.

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) If a cat's purr is about 25 dB, approximately how many times as intense is a cat's purr relative to the threshold of sound? **2**

.....

.....

.....

.....

.....

.....

.....

.....

.....

Questions 11 – 37 are worth 75 marks in total
--

Student Number: _____

Question 38 (7 marks)

A particle is moving along the x -axis. Its displacement x centimetres at time t seconds is given by $x = 2t^3 - 5t^2 - 2t + 1$ for $t \geq 0$.

- (a) What is the initial displacement of the particle? 1

.....

.....

.....

- (b) What is the particle’s initial velocity? 2

.....

.....

.....

.....

.....

.....

- (c) How many times after $t = 0$ is the particle stationary? Justify your answer with calculations. 2

.....

.....

.....

.....

.....

.....

- (d) Calculate the average velocity of the particle during the first two seconds. 2

.....

.....

.....

.....

.....

4

Diagram of a triangular plot ABC with a point D on side AC . A line segment BD is drawn. The length of BD is 48 m , the length of BC is 50 m , and the angle BDC is 28° . The length of AC is 120 m .

[illegible]

Question 40 (4 marks)

Solve the trigonometric equation $2 \sin^2 2x - \sin 2x = 1$ for $0 \leq x \leq 2\pi$.

4

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

End of paper

STUDENT NUMBER: _____



YEAR 11 MATHEMATICS ADVANCED
2020 YEAR 11 ASSESSMENT TASK 3

Answer Sheet for Section I

- | | | | | |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]



2020 YEAR 11 ASSESSMENT TASK 3

Mathematics Advanced

Thursday 10th September

General**Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
100**Section I – 10 marks**

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks

- Attempt Questions 11– 40
- Allow about 2 hours and 45 minutes for this section

Name: Solutions

Student Number: _____

Outcomes

A student:

- MA11-1 uses algebraic and graphical techniques to solve, and where appropriate, compare alternative solutions to problems
- MA11-2 uses the concepts of functions and relations to model, analyse and solve practical problems
- MA11-3 uses the concepts and techniques of trigonometry in the solution of equations and problems involving geometric shapes
- MA11-4 uses the concepts and techniques of periodic functions in the solutions of trigonometric equations or proof of trigonometric identities
- MA11-5 interprets the meaning of the derivative, determines the derivative of functions and applies these to solve simple practical problems
- MA11-6 manipulates and solves expressions using the logarithmic and index laws, and uses logarithms and exponential functions to solve practical problems
- MA11-7 uses concepts and techniques from probability to present and interpret data and solve problems in a variety of contexts, including the use of probability distributions
- MA11-9 provides reasoning to support conclusions which are appropriate to the context

Section I

10 marks

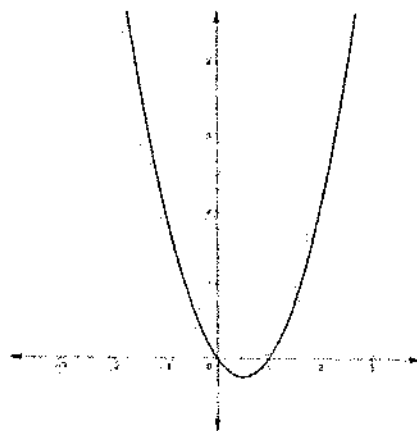
Attempt Questions 1–10

Allow about 15 minutes for this section

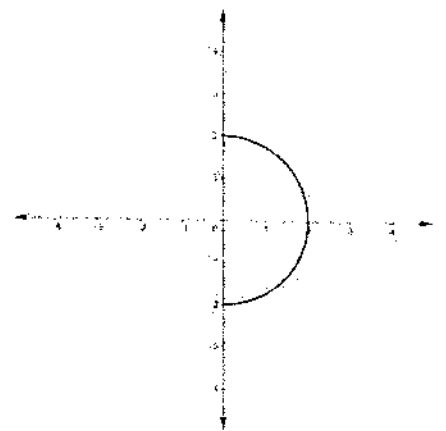
Use the multiple-choice answer sheet for Questions 1–10

1 Which graph represents a one-to-one function?

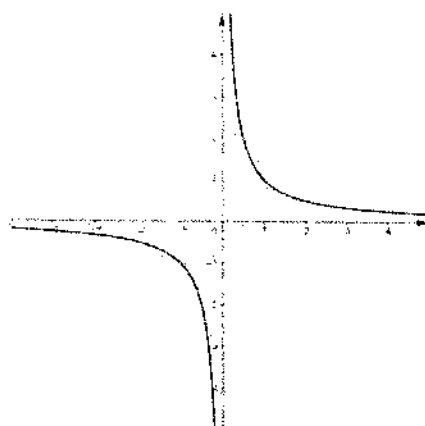
A.



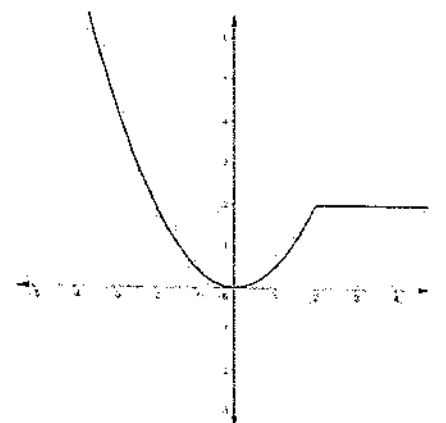
B.



C.



D.



- 2 If $f(x) = x^2 - 2x$, what is the value of $f(-3)$?

- A. -15
B. -3
C. 0
☒ D. 15

$$\begin{aligned} f(-3) &= (-3)^2 - 2(-3) \\ &= 9 + 6 \\ &= 15 \end{aligned}$$

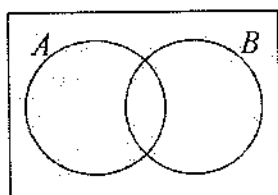
- 3 How many x -intercepts does the graph of $f(x) = (x-2)(x^2 + 4x + 3)$ have?

- A. 0
B. 1
C. 2
☒ D. 3

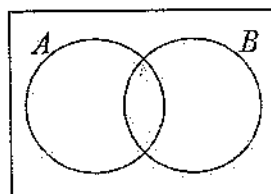
$$= (x-2)(x-3)(x+1)$$

- 4 In which of the Venn diagrams is $A \cap B'$ shaded?

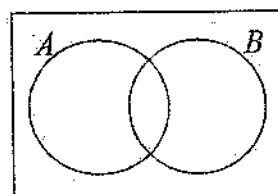
☒ A.



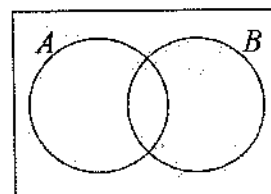
B.



C.



D.



- 5 Which of the following lines is parallel to the line $6x - 2y - 3 = 0$?

A. $6x - y - 5 = 0$

☒ B. $y = 3x - 5$

C. $y = \frac{1}{3}x - 8$

D. $y = -3x + 5$

$$6x - 3 = 2y$$

$$3x - \frac{3}{2} = y$$

$$\therefore m = 3$$

6 What are the solutions to $|2x+3|=1$?

☒ A. $x=-2$ or $x=-1$

B. $x=-2$ or $x=1$

C. $x=1$ or $x=2$

D. $x=2$ or $x=-1$

$$2x+3=1 \quad \text{or} \quad -(2x+3)=1$$

$$2x=-2$$

$$x=-1$$

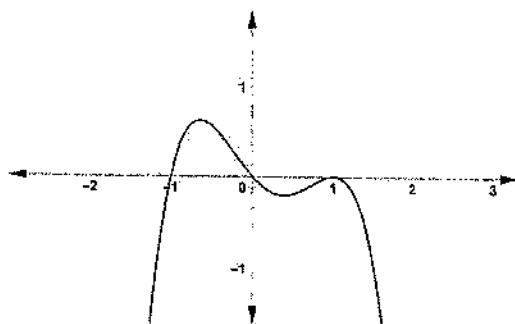
$$2x+3=-1$$

$$2x=-4$$

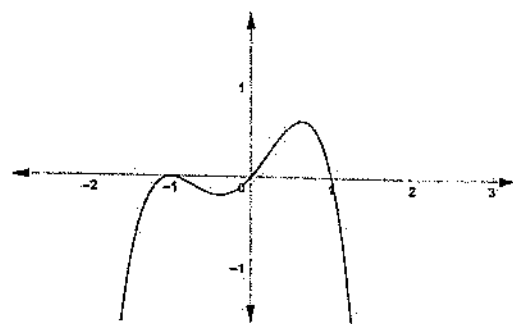
$$x=-2$$

7 Which of the following graphs shows the function $y=-x(x-1)^2(x+1)$?

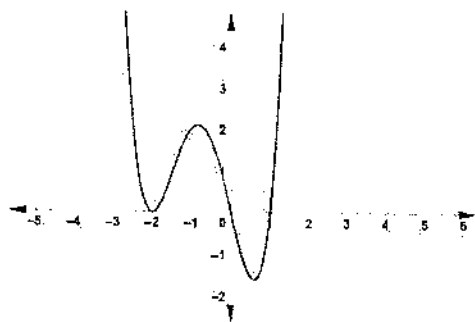
☒ A.



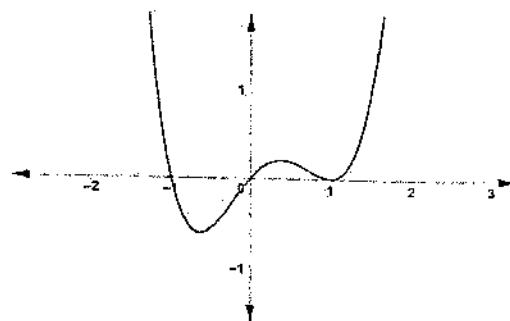
B.



C.



D.



8 Which of the following is always true for $0^\circ < \theta < 90^\circ$?

A. $\sin(90-\theta) = \sin \theta$

☒ B. $\sin(180-\theta) = \sin \theta$

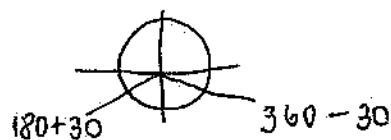
C. $\sin(180+\theta) = \sin \theta$

D. $\sin(360-\theta) = \sin \theta$

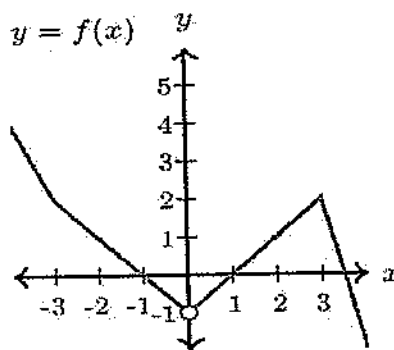
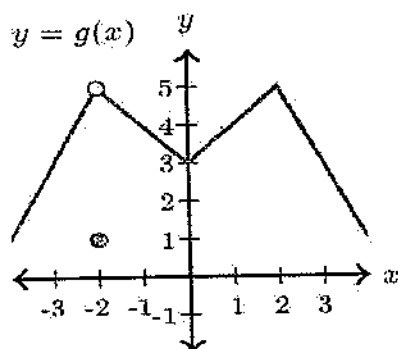
- 9 What are the solutions to $\sin x = -\frac{1}{2}$ for $0 \leq x \leq 360^\circ$?

- A. $x = 30^\circ$
 B. $x = 30^\circ$ and $x = 150^\circ$
 C. $x = 150^\circ$ and $x = 210^\circ$
 D. $x = 210^\circ$ and $x = 330^\circ$

$$\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$$



- 10 The functions $f(x)$ and $g(x)$ are graphed below.
 What is the value of $g(f(x))$ when $x = -2$?



- A. 0
 B. 1
 C. 4
 D. The value does not exist.

$$f(-2) = 1$$

$$g(1) = 4$$

STUDENT NUMBER:

Solutions



YEAR 11 MATHEMATICS ADVANCED
2020 YEAR 11 ASSESSMENT TASK 3

Answer Sheet for Section I

- | | | | | |
|----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 | A ☐ | B ☐ | C ☒ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☒ |
| 3 | A ☐ | B ☐ | C ☐ | D ☒ |
| 4 | A ☒ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☒ | C ☐ | D ☐ |
| 6 | A ☒ | B ☐ | C ☐ | D ☐ |
| 7 | A ☒ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☒ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☒ |
| 10 | A ☐ | B ☐ | C ☒ | D ☐ |

Student Number: Solutions

2020 YEAR 11 ASSESSMENT TASK 3

Mathematics Advanced

Section II Answer Booklet

90 marks

Attempt Questions 11 - 40

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing paper is provided at the back of each booklet. If you use this space, clearly indicate which question you are answering.
-

Question 11 (2 marks)

Simplify $\frac{2x^2 - 5x - 12}{x - 4}$.

2

$$= \frac{(2x + 3)(x - 4)}{x - 4} \quad 1 \quad \textcircled{2} \text{ correct solution}$$

$$= 2x + 3 \quad 1 \quad \textcircled{1} \text{ substantially correct solution}$$

Question 12 (3 marks)

Write $\frac{20}{x^2 + 5x} - \frac{4}{x}$ as a single fraction in simplest form.

3

$$= \frac{20}{x(x+5)} - \frac{4}{x} \quad \textcircled{3} \text{ correct solution fully simplified}$$

$$= \frac{20 - 4(x+5)}{x(x+5)} \quad 1 \quad \textcircled{2} \text{ substantially correct solution - correct expansion and collection of like terms}$$

$$= \frac{20 - 4x - 20}{x(x+5)} \quad \textcircled{1} \text{ correctly expressing fraction with a common denominator}$$

$$= \frac{-4x}{x(x+5)} \quad 1$$

$$= \frac{-4}{x+5}$$

Question 13 (1 mark)

Evaluate $\log_a a^2$.

1

$$= 2$$

Question 14 (2 marks)

Solve $\log_2(x+2) = 3$.

2

$$\begin{aligned} \log_2(x+2) &= 3 \\ 2^3 &= x+2 & 1 \\ 8 &= x+2 \\ \therefore x &= 6 & 1 \end{aligned}$$

(2) correct solution

(1) Rewriting equation without log.

Question 15 (1 mark)

Convert 225° to radians in exact form.

1

$$\begin{aligned} 225 \times \frac{\pi}{180} \\ = \frac{5\pi}{4} \end{aligned}$$

(1) correct answer

Question 16 (2 marks)

State the natural domain of $f(x) = \frac{5}{\sqrt{5-x}}$ using interval notation.

2

$$\begin{aligned} 5-x &> 0 \\ 5 &> x & \therefore x < 5 \end{aligned}$$

(2) correct domain

(1) Realising $x \neq 5$
or $x < 5$

$$D: x \in (-\infty, 5)$$

Question 17 (2 marks)

Determine whether the function $f(x) = x^3 + 2x$ is odd, even or neither and justify your answer.

2

$$\begin{aligned} f(-x) &= (-x)^3 + 2(-x) \\ &= -x^3 - 2x \\ &= -(x^3 + 2x) \\ &= -f(x) \end{aligned}$$

② correct solution with conclusion

① substantially correct solution

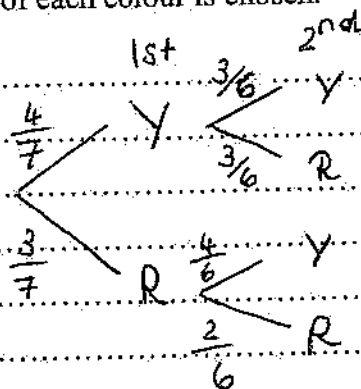
$\therefore f(x)$ is an odd function

or
correctly finding $f(-x)$ simplified

Question 18 (2 marks)

A bag contains 4 yellow and 3 red balls. Two balls are chosen at random and the first ball is not replaced before the second ball is chosen. Find the probability that one ball of each colour is chosen.

2



② correct solution

① correct prob. tree

or correct calculation from incorrect interpretation.

$$P(RY) + P(YR) = \frac{3}{7} \times \frac{4}{6} + \frac{4}{7} \times \frac{3}{6}$$

$$= \frac{24}{42}$$

$$= \frac{4}{7}$$

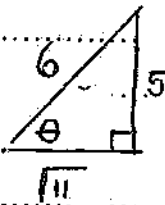
② correct solution

① correct ratio for acute angle θ .

Question 19 (2 marks)

Given $\sin \theta = \frac{5}{6}$ and $90^\circ \leq \theta \leq 180^\circ$, find the exact value of $\cos \theta$.

2



$$6^2 = 5^2 + x^2$$

$$x^2 = 11$$

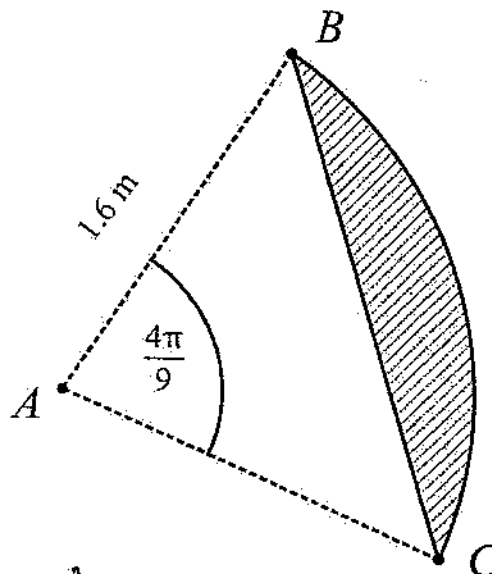
$$x = \sqrt{11}, x > 0$$

Since θ is obtuse $\cos \theta = -\frac{\sqrt{11}}{6}$

Question 20 (2 marks)

The shaded segment shown is formed by an arc BC of a circle with radius 1.6 metres and the chord BC which subtends an angle of $\frac{4\pi}{9}$ radians at its centre A . Find the area of the shaded segment, correct to 3 significant figures.

2



$$A = \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta$$

$$= \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} (1.6)^2 \left(\frac{4\pi}{9} - \sin \frac{4\pi}{9} \right)$$

$$= 0.52666 \dots$$

$$= 0.527 \text{ m}^2 \text{ (3 sig. fig.)}$$

② correct solution
correctly rounded

① substantially
correct solution

Questions 11 – 20 are worth 19 marks in total

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

F 10
C 5
P 4
J 4

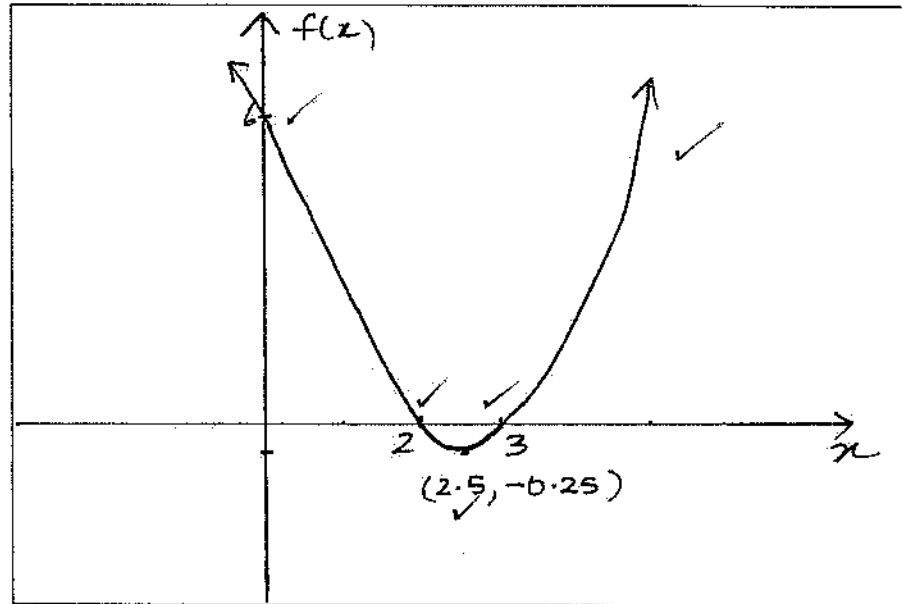
Student Number: _____

Question 21 (4 marks)

- (a) Sketch the graph of the function $f(x) = x^2 - 5x + 6$ showing all intercepts and important features.

2 F

$f(x) = x^2 - 5x + 6$
 $= (x-2)(x-3)$
 $\therefore x$ -intercepts $x=3, 2$
 y -intercept $y=6$
 $\text{vertex } x = -\frac{(-5)}{2}$
 $= 2.5$
 $y = -0.25$
 $\therefore (2.5, -0.25)$



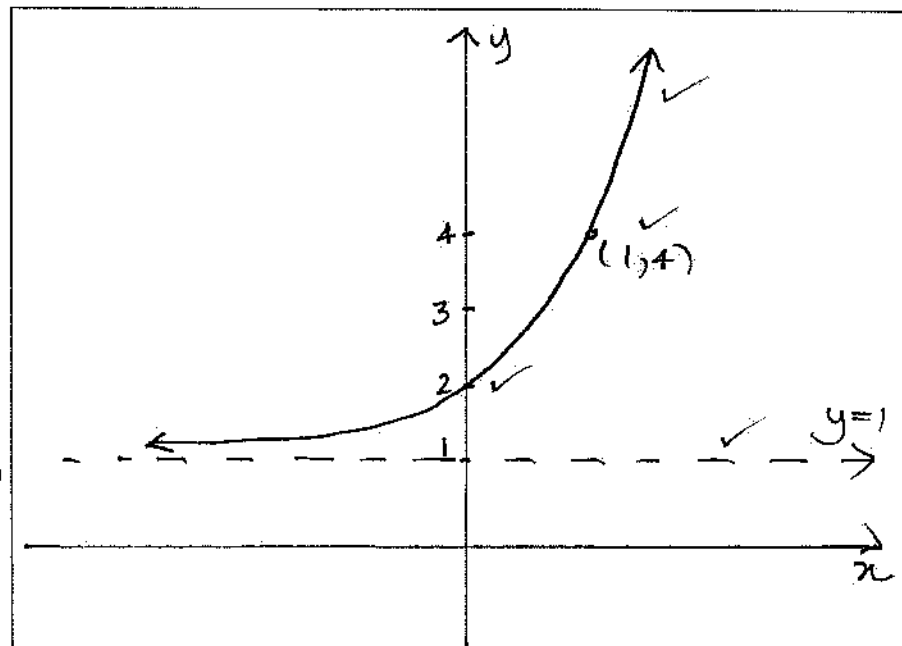
correct graph including x, y intercepts, vertex and shape.

① substantially correct graph

- (b) Sketch the graph of the function $f(x) = 1 + 3^x$ showing all important features.

2 F

$f(x) = 1 + 3^x$
 $\therefore$ graph of $y = 3^x$
 translated up 1.



② correct graph with asymptote, y intercept, point on curve, shape.

① attempt to translate $y = 3^x$ up 1.

Question 22 (4 marks)

The table shows the value of tips Isabella receives when she works in a restaurant on a Friday night. The probabilities of each value being received are based on experience.

Tips (to the nearest Dollar)	\$5	\$12	\$23	\$35	\$38	\$40
Probability	0.15	0.26	0.3	0.16	0.1	0.03

- (a) Explain why the above table is a probability distribution.

② J.

② correct statement

① correct statement

$$\begin{aligned} & \bullet 0 \leq P(X=x) \leq 1 \\ & \bullet \sum p x_i = 0.15 + 0.26 + 0.3 + 0.16 + 0.1 + 0.03 \\ & \quad = 1 \end{aligned}$$

The probabilities are all from 0 to 1, and the sum of the probabilities is one.

- (b) Calculate the expected value of Isabella's tips for the next Friday she works.

1

$$\begin{aligned} E(X) &= 5 \times 0.15 + 12 \times 0.26 + 23 \times 0.3 + 35 \times 0.16 \\ &\quad + 38 \times 0.1 + 40 \times 0.03 \\ &= \$21.37 \end{aligned}$$

- (c) What is the percentage probability that Isabella will receive at least \$35 worth of tips?

1

$$\begin{aligned} P(\text{at least } 35) &= P(X=35) + P(X=38) + P(X=40) \\ &= 0.16 + 0.1 + 0.03 \\ &= 0.29 \\ &= 29\% \end{aligned}$$

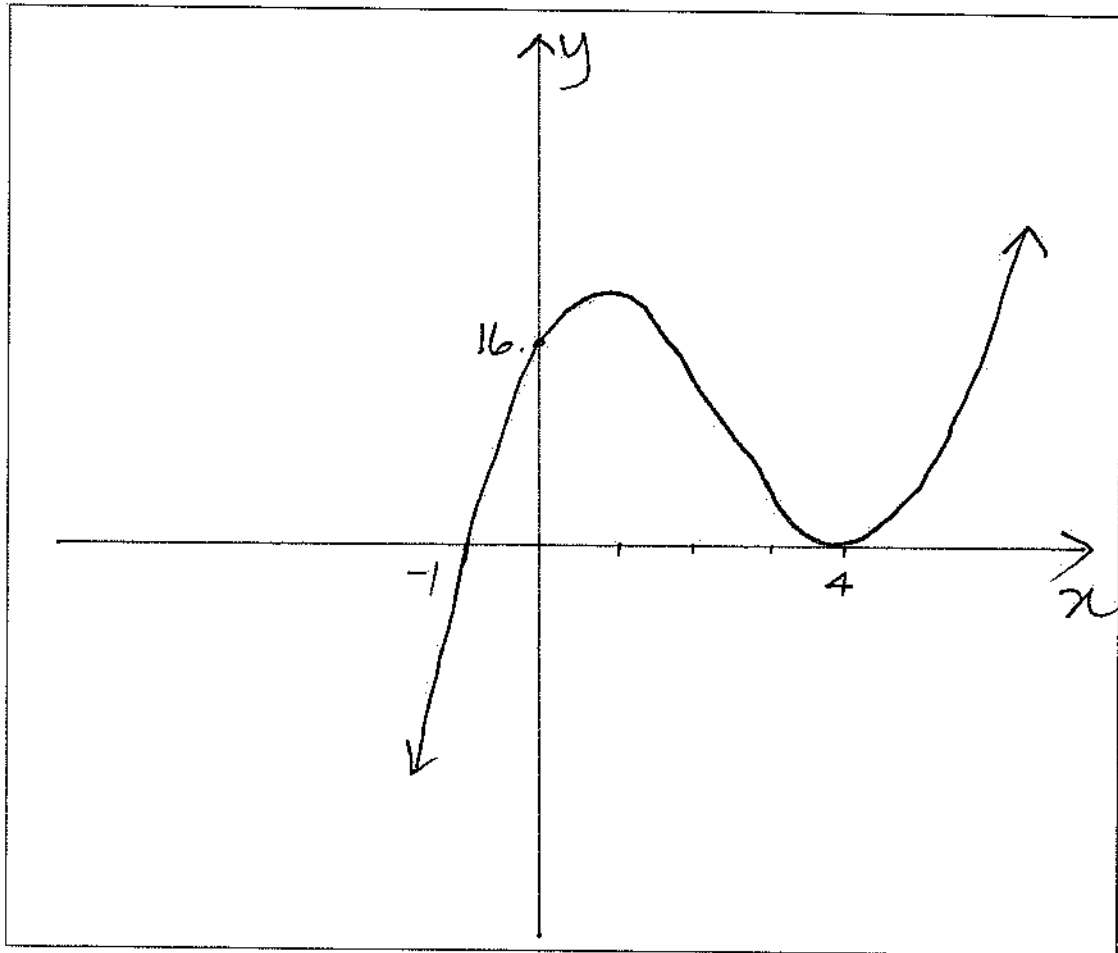
Question 23 (2 marks)

Sketch the graph of $y = (x-4)^2(x+1)$.

double root at $x=4$

single root at $x=-1$

2



$$\text{when } x=0, \quad y = (-4)^2(1) \\ = 16.$$

Question 24 (4 marks)

- a) Find the centre and the radius of the circle represented by the equation.

2

$$x^2 - 6x + y^2 + 2y + 1 = 0$$

$$(x^2 - 6x) + (y^2 + 2y) = -1$$

$$(x^2 - 6x + 9) + (y^2 + 2y + 1) = -1 + 9 + 1$$

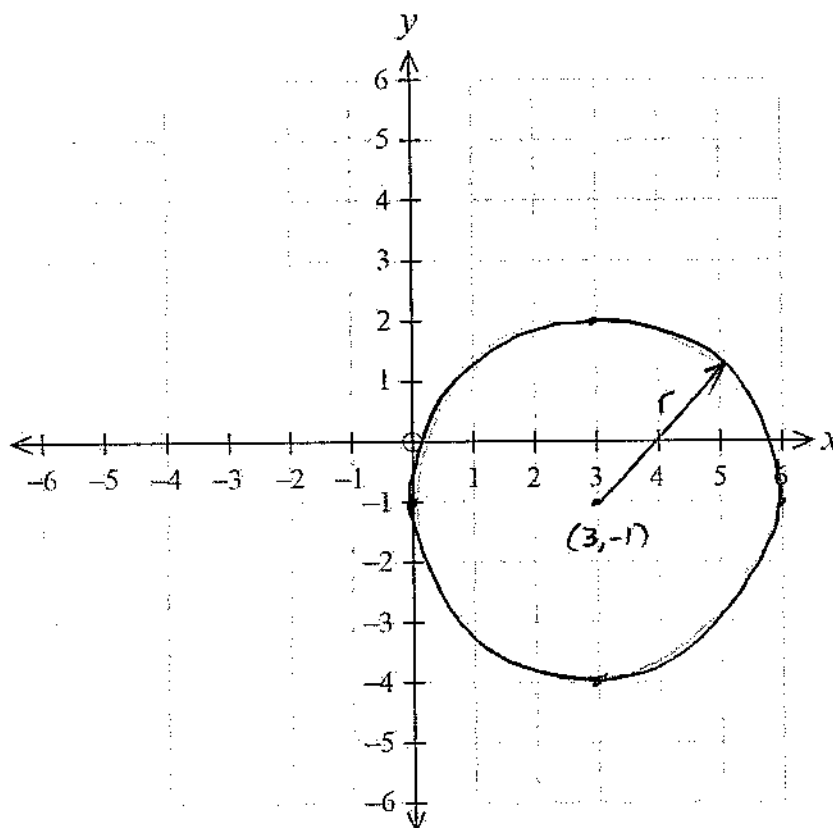
$$(x - 3)^2 + (y + 1)^2 = 9$$

Centre (3, -1)

Radius 3

- (b) Draw the circle on the number plane below. You do **not** need to find any x-intercepts.

2



Question 25 (5 marks).

- (a) Find the equation of the tangent to the curve $y = 3x^3 - 10x^2 + 1$ at the point where $x = 2$.

3

③ correct solution

$$y = 3x^3 - 10x^2 + 1$$

$$\frac{dy}{dx} = 9x^2 - 20x$$

② correct derivative and y. value or equivalent.

When $x = 2$, $y = 3(2)^3 - 10(2)^2 + 1$
 $= -15$

$$m_t = 9(2)^2 - 20(2)$$

$$= -4$$

∴ Gradient tangent at $(2, -15)$ is -4 .

Equation of tangent.

① correct derivative or y. value.

$$y - y_1 = m(x - x_1)$$

$$y + 15 = -4(x - 2)$$

$$y + 15 = -4x + 8$$

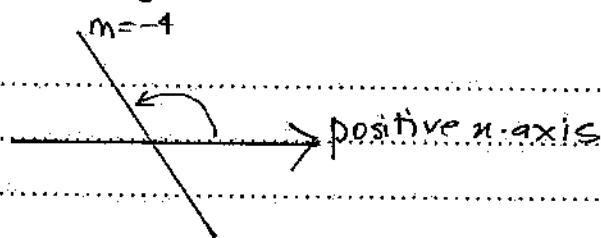
④ Use of words to aid in setting/ understanding

$$y = -4x - 7 \quad \text{or} \quad 4x + y + 7 = 0.$$

- (b) Find the angle of inclination between the tangent and the positive x-axis to the nearest degree.

2

② correct angle.



$$\tan \theta = m$$

$$\tan \theta = -4$$

$$\theta = 180 - \tan^{-1}(4)$$

$$= 104.0362$$

$$\theta = 104^\circ (\text{nearest } ^\circ)$$

① finding acute angle or equivalent

∴ The angle of inclination is 104° .

Justification 1 obtuse angle.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

1. Introduction
 2. Background
 3. Methodology
 4. Results
 5. Discussion
 6. Conclusion
 7. References
 8. Appendix
 9. Figure 1
 10. Figure 2
 11. Figure 3
 12. Figure 4
 13. Figure 5
 14. Figure 6
 15. Figure 7
 16. Figure 8
 17. Figure 9
 18. Figure 10
 19. Figure 11
 20. Figure 12
 21. Figure 13
 22. Figure 14
 23. Figure 15
 24. Figure 16
 25. Figure 17
 26. Figure 18
 27. Figure 19
 28. Figure 20
 29. Figure 21
 30. Figure 22
 31. Figure 23
 32. Figure 24
 33. Figure 25
 34. Figure 26
 35. Figure 27
 36. Figure 28
 37. Figure 29
 38. Figure 30
 39. Figure 31
 40. Figure 32
 41. Figure 33
 42. Figure 34
 43. Figure 35
 44. Figure 36
 45. Figure 37
 46. Figure 38
 47. Figure 39
 48. Figure 40
 49. Figure 41
 50. Figure 42
 51. Figure 43
 52. Figure 44
 53. Figure 45
 54. Figure 46
 55. Figure 47
 56. Figure 48
 57. Figure 49
 58. Figure 50
 59. Figure 51
 60. Figure 52
 61. Figure 53
 62. Figure 54
 63. Figure 55
 64. Figure 56
 65. Figure 57
 66. Figure 58
 67. Figure 59
 68. Figure 60
 69. Figure 61
 70. Figure 62
 71. Figure 63
 72. Figure 64
 73. Figure 65
 74. Figure 66
 75. Figure 67
 76. Figure 68
 77. Figure 69
 78. Figure 70
 79. Figure 71
 80. Figure 72
 81. Figure 73
 82. Figure 74
 83. Figure 75
 84. Figure 76
 85. Figure 77
 86. Figure 78
 87. Figure 79
 88. Figure 80
 89. Figure 81
 90. Figure 82
 91. Figure 83
 92. Figure 84
 93. Figure 85
 94. Figure 86
 95. Figure 87
 96. Figure 88
 97. Figure 89
 98. Figure 90
 99. Figure 91
 100. Figure 92
 101. Figure 93
 102. Figure 94
 103. Figure 95
 104. Figure 96
 105. Figure 97
 106. Figure 98
 107. Figure 99
 108. Figure 100
 109. Figure 101
 110. Figure 102
 111. Figure 103
 112. Figure 104
 113. Figure 105
 114. Figure 106
 115. Figure 107
 116. Figure 108
 117. Figure 109
 118. Figure 110
 119. Figure 111
 120. Figure 112
 121. Figure 113
 122. Figure 114
 123. Figure 115
 124. Figure 116
 125. Figure 117
 126. Figure 118
 127. Figure 119
 128. Figure 120
 129. Figure 121
 130. Figure 122
 131. Figure 123
 132. Figure 124
 133. Figure 125
 134. Figure 126
 135. Figure 127
 136. Figure 128
 137. Figure 129
 138. Figure 130
 139. Figure 131
 140. Figure 132
 141. Figure 133
 142. Figure 134
 143. Figure 135
 144. Figure 136
 145. Figure 137
 146. Figure 138
 147. Figure 139
 148. Figure 140
 149. Figure 141
 150. Figure 142
 151. Figure 143
 152. Figure 144
 153. Figure 145
 154. Figure 146
 155. Figure 147
 156. Figure 148
 157. Figure 149
 158. Figure 150
 159. Figure 151
 160. Figure 152
 161. Figure 153
 162. Figure 154
 163. Figure 155
 164. Figure 156
 165. Figure 157
 166. Figure 158
 167. Figure 159
 168. Figure 160
 169. Figure 161
 170. Figure 162
 171. Figure 163
 172. Figure 164
 173. Figure 165
 174. Figure 166
 175. Figure 167
 176. Figure 168
 177. Figure 169
 178. Figure 170
 179. Figure 171
 180. Figure 172
 181. Figure 173
 182. Figure 174
 183. Figure 175
 184. Figure 176
 185. Figure 177
 186. Figure 178
 187. Figure 179
 188. Figure 180
 189. Figure 181
 190. Figure 182
 191. Figure 183
 192. Figure 184
 193. Figure 185
 194. Figure 186
 195. Figure 187
 196. Figure 188
 197. Figure 189
 198. Figure 190
 199. Figure 191
 200. Figure 192
 201. Figure 193
 202. Figure 194
 203. Figure 195
 204. Figure 196
 205. Figure 197
 206. Figure 198
 207. Figure 199
 208. Figure 200
 209. Figure 201
 210. Figure 202
 211. Figure 203
 212. Figure 204
 213. Figure 205
 214. Figure 206
 215. Figure 207
 216. Figure 208
 217. Figure 209

Student Number: Solutions

Question 26 (2 marks)

At a Performing Arts College 70% of students study Drama, 55% of students study Music and 35% of students study both Drama and Music.

- (a) What is the probability that a randomly selected student studies Music or Drama or both?

1

$$\begin{aligned} P(M \cup D) &= P(M) + P(D) - P(M \cap D) \\ &= 70\% + 55\% - 35\% \\ &= 90\% \end{aligned}$$

① correct solution

- (b) Given that a student studies Drama what is the probability that this student also studies Music?

1

$$\begin{aligned} P(M|D) &= \frac{P(M \cap D)}{P(D)} \\ &= \frac{0.35}{0.7} \\ &= 0.5 \end{aligned}$$

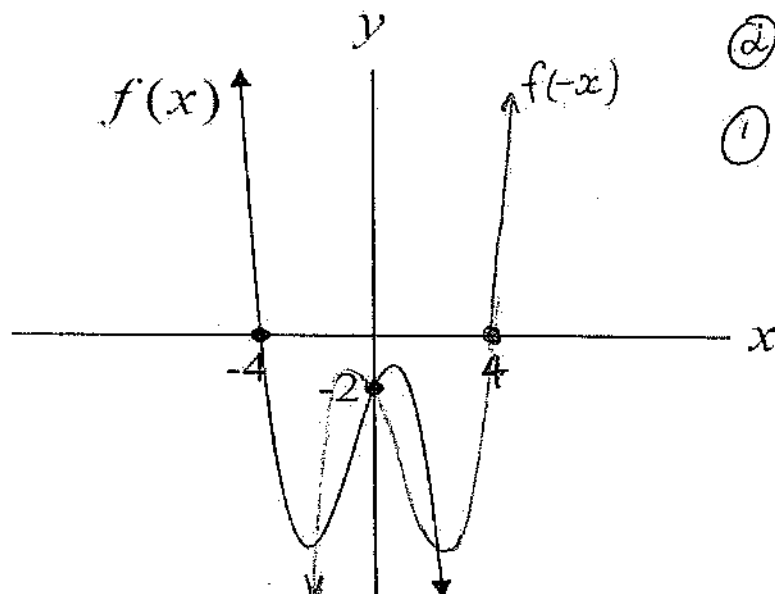
① correct sol.

Question 27 (2 marks)

The diagram shows the curve $y = f(x)$.

Sketch $y = f(-x)$ on the same set of axes, showing any x - and y -intercepts.

2



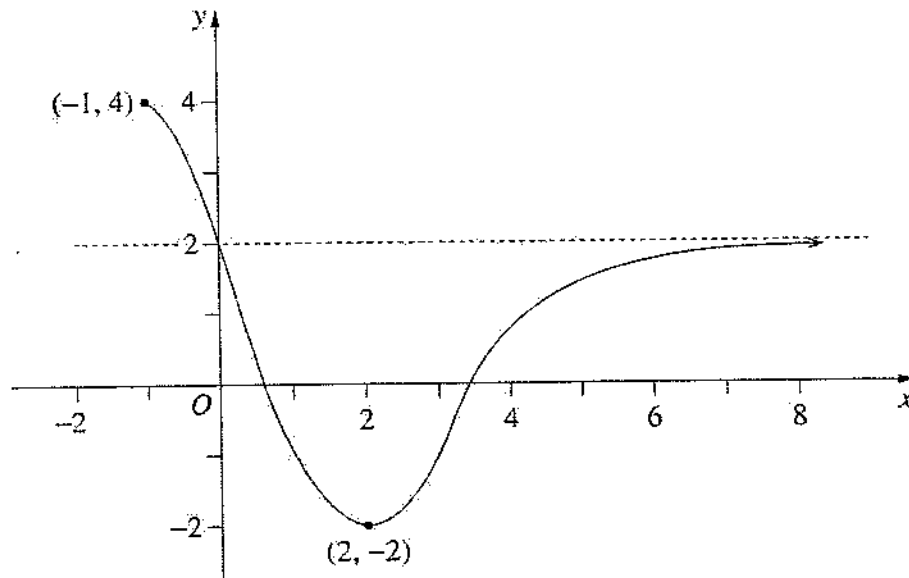
② correct graph

① subst. correct graph

Question 28 (2 marks)

The graph of a function $y = f(x)$ is shown. It has an asymptote at $y = 2$.

2



Using interval notation, state the domain and range of $y = f(x)$.

Domain $x \in [-1, \infty)$

Range $y \in [-2, 4]$

② correct domain and range

① either domain or range correct

Question 29 (8 marks)

Differentiate the following with respect to x

(a) $y = 2x(x+7)$

1

$= 2x^2 + 14x$ ① correct derivative

$y' = 4x + 14$

(b) $y = 3(x^2-1)^3$

2

$y' = 9(x^2-1)^2 \times 2x$ ② correct derivative (simplified)

$= 18x(x^2-1)^2$

① substantially correct derivative)

(c) $y = 2x\sqrt{x}$

2

$= 2x^{3/2}$

② correct derivative
① correctly using index

$y' = 2 \times \frac{3}{2} x^{1/2}$

$= 3\sqrt{x}$ or $3x^{1/2}$

(d) $y = \frac{2x^3}{x^2-5}$

3

$u = 2x^3$ ~~$v = x^2-5$~~

$u' = 6x^2$ ~~$v' = 2x$~~

$y' = \frac{vu' - uv'}{v^2}$

$= \frac{6x^2(x^2-5) - 4x^4}{(x^2-5)^2}$ ③ correct derivative

$= \frac{6x^4 - 30x^2 - 4x^4}{(x^2-5)^2}$ ② correct subst. into form

$= \frac{2x^4 - 30x^2}{(x^2-5)^2}$ ① correct differentiation of num. and denom.

Question 30 (2 marks)

A piecewise function $f(x)$ is defined as follows.

2

$$f(x) = \begin{cases} -x + a, & x < 1 \\ x^2 - ax + 2, & x \geq 1 \end{cases}$$

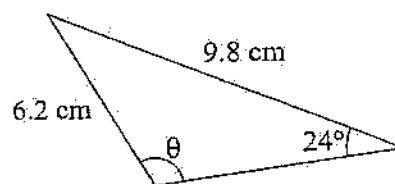
What value of a makes this function continuous?

$$\begin{aligned} -x + a &= x^2 - ax + 2 \text{ when } x = 1 \\ -1 + a &= 1^2 - a + 2 \\ -1 + a &= 3 - a \quad (2) \\ 2a &= 4 \\ a &= 2 \quad (1) \end{aligned}$$

Question 31 (3 marks)

The diagram shows an obtuse triangle with side lengths 6.2 cm and 9.8 cm, and angles 24° and θ , where θ is the largest angle.

3



Find the value of θ to the nearest degree.

$$\begin{aligned} \frac{\sin \theta}{9.8} &= \frac{\sin 24^\circ}{6.2} && \text{But } \theta \text{ needs to be obtuse, so} \\ \sin \theta &= \frac{9.8 \sin 24^\circ}{6.2} && \theta = 180 - 40.0088 \dots \\ & && = 140^\circ \text{ (nearest deg.)} \\ \theta &= \sin^{-1} \left(\frac{9.8 \sin 24^\circ}{6.2} \right) && (3) \text{ correct solution} \\ & && (2) \text{ correct acute angle} \\ &= 40.0088 \dots^\circ && (1) \text{ correct substitution into sine rule} \end{aligned}$$

Question 32 (3 marks)

The function $f(x) = 2x(x-4)^3$ has two horizontal tangents. Given that one horizontal tangent touches the curve at $(4, 0)$, find the coordinates of the other point on the curve at which the tangent is horizontal.

3

$$f(x) = 2x(x-4)^3$$

$$u = 2x \quad v = (x-4)^3$$

$$u' = 2 \quad v' = 3(x-4)^2$$

$$f'(x) = vu' + uv'$$

$$= 2(x-4)^3 + 6x(x-4)^2$$

$$= 2(x-4)^2(x-4 + 3x)$$

$$= 2(x-4)^2(4x-4)$$

$$= 8(x-4)^2(x-1)$$

$$f'(x) = 0 \text{ at } x = 4, x = 1$$

$$\text{When } x = 1 \quad f(x) = 2(1)(1-4)^3$$

$$= -54$$

∴ Horizontal tangents at $(4, 0)$ and $(1, -54)$

③ correct solution

② correct solving of $f'(x) = 0$

① correct derivative

Question 33 (3 marks)

3

Solve $2\cos^2 x - \cos x = 0$ for $0 \leq x \leq 2\pi$.

$$\cos x (2\cos x - 1) = 0$$

$$\cos x = 0 \quad \text{or} \quad 2\cos x - 1 = 0$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \quad x = \frac{\pi}{3}, \frac{5\pi}{3}$$

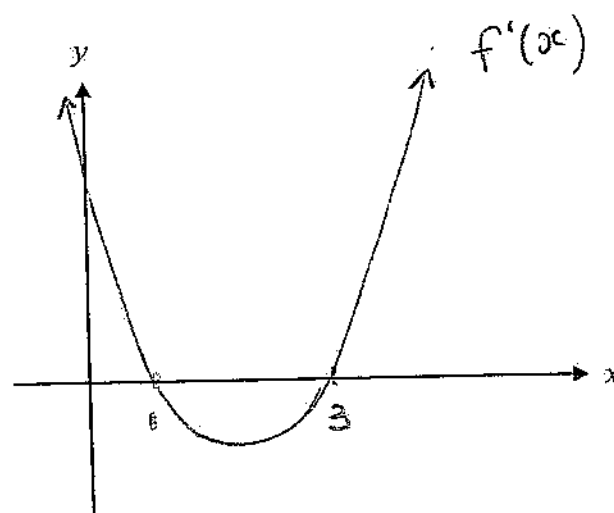
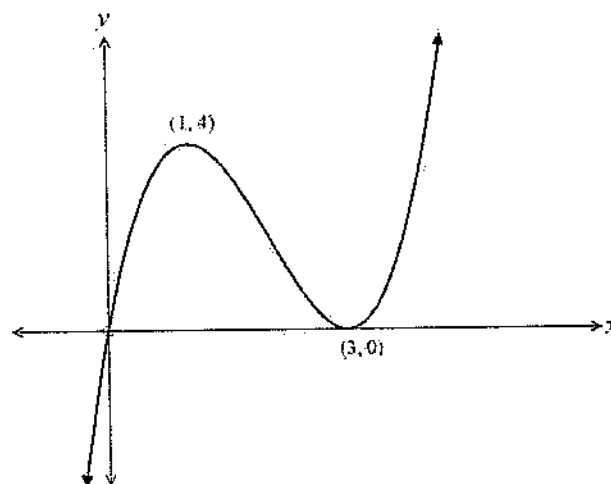
$$\therefore x = \frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{3}$$

③ correct solution
② correctly finding acute solutions
① correct factorisation

Question 34 (2 marks)

The diagram shows the graph of the curve $y = f(x)$. Sketch the graph of the gradient function on the number plane below.

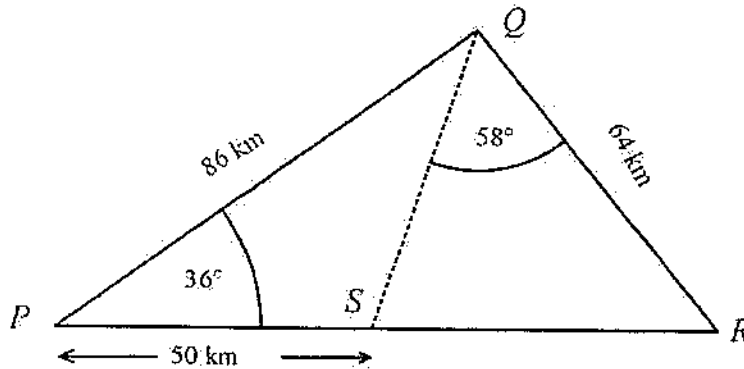
2



- ② correct shape and intercepts
① correct shape

Question 35 (4 marks)

Triangles PQS and QRS have a common side QS , as shown.



- (a) Show that the length $QS = 54.2$ km correct to 3 significant figures.

2

$$\begin{aligned} (QS)^2 &= 50^2 + 86^2 - 2(50)(86) \cos 36^\circ \\ &= 2938.4538 \\ QS &= 54.20750 \dots \quad QS > 0 \\ &= 54.2 \quad (3 \text{ s.f.}) \end{aligned}$$

② correct length (unrounded)

① correct substitution in formula

- (b) Hence or otherwise find the area of ΔPQR correct to the nearest square kilometre.

2

$$\begin{aligned} \text{Area } PQR &= \frac{1}{2} \times 86 \times 50 \sin 36^\circ + \frac{1}{2} \times 64 \times 54.2 \sin 58^\circ \\ &= 2734.592 \\ &= 2735 \text{ m}^2 \quad (\text{nearest m}^2) \end{aligned}$$

Also accept:

$$A = 2728 \text{ km}^2$$

or

$$A = 2752 \text{ km}^2$$

with correct working.

② correct area

① finding one area correctly

Question 36 (2 marks)

Use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ given $f(x) = x^2 - 2x$.

2

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[(x+h)^2 - 2(x+h)] - (x^2 - 2x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 2h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 2)}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h - 2$$

$$= 2x - 2$$

(2) correct differentiation by first principles

(1) substantially correct solution

Question 37 (4 marks)

The decibel, dB, is a logarithmic unit used to measure sound level. The formula to find the decibel level of a sound is $\text{dB} = 10 \log_{10} \left(\frac{I}{I_0} \right)$ where I is the intensity of the sound, and I_0 is the intensity of the threshold of sound (sound which can barely be perceived).

- (a) According to Safe Work Australia, workers should wear ear protection if they are exposed to sound levels of 85 dB or more for an extended period of time. A machine produces sound at an intensity of $(1.8 \times 10^{10}) I_0$. Should workers who spend time near this machine wear ear protection? Justify your answer with calculations. 2

$$\begin{aligned} \text{dB} &= 10 \log_{10} \left(\frac{I}{I_0} \right) \\ &= 10 \log_{10} \left(\frac{1.8 \times 10^{10} I_0}{I_0} \right) \quad (2) \\ &= 10 \log_{10} (1.8 \times 10^{10}) \quad (1) \\ &= 102.5527 \dots \\ &\approx 103 \text{ dB} \end{aligned}$$

$\therefore$ Yes, they should wear protection.

- (b) If a cat's purr is about 25 dB, approximately how many times as intense is a cat's purr relative to the threshold of sound? 2

$$\begin{aligned} 25 &= 10 \log_{10} \left(\frac{I}{I_0} \right) \\ 2.5 &= \log_{10} \left(\frac{I}{I_0} \right) \\ 10^{2.5} &= \frac{I}{I_0} \\ \frac{I}{I_0} &= 316.2277 \dots \end{aligned}$$

$\therefore$ It is approximately 316 times the threshold of sound.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[This page contains horizontal dotted lines for writing.]

Student Number: Solution

Question 38 (7 marks)

A particle is moving along the x -axis. Its displacement x centimetres at time t seconds is given by $x = 2t^3 - 5t^2 - 2t + 1$ for $t \geq 0$.

- (a) What is the initial displacement of the particle?

1

$t = 0$ $x = 1$ ✓
 $\therefore 1$ cm to the right of the origin

- (b) What is the particle's initial velocity?

2

$\dot{v} = 6t^2 - 10t - 2$ ✓ (2) correct solution
 when $t = 0$ (1) correct derivative
 $v = -2$ cm/s ✓
 $\therefore$ The particle is moving to the left at 2 cm/sec.
 or -2 cm/s

- (c) How many times after $t = 0$ is the particle stationary? Justify your answer with calculations.

2

Stationary when $v = 0$ (2) correct sol.
 $6t^2 - 10t - 2 = 0$ $t_1 = 1.847$ (1) only once
 $3t^2 - 5t - 1 = 0$ $t_2 = -0.180$
 $t = \frac{5 \pm \sqrt{(-5)^2 - 4(3)(-1)}}{2 \times 3}$
 $= \frac{5 \pm \sqrt{37}}{6}$ Since $t \geq 0$, the particle is stationary only once after $t = 0$

- (d) Calculate the average velocity of the particle during the first two seconds.

2

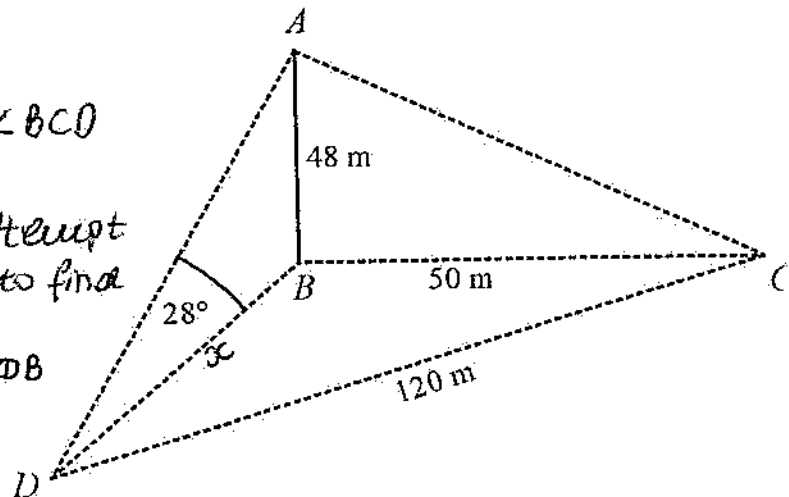
Average velocity $= \frac{x_2 - x_1}{t_2 - t_1}$ (2) correct sol.
 $x(2) = -7$
 $x(0) = 1$
 $= \frac{-7 - 1}{2 - 0}$ (1) correctly finding $x(2) = -7$
 $= -4$ cm/sec
 $= -8$ cm/sec
 $\therefore$ Average velocity 8 cm/sec to the left.

Question 39 (4 marks)

4

A 48-metre high, vertical pole AB stands in the middle of a flat sporting ground. Diana starts from the base of the pole and walks 50 m due east to a point C where she places a peg. She attaches a 120 m long string to the peg and walks in a roughly south-westerly direction until the string is taut, at point D . She then measures the angle of elevation of the pole to be 28° .

④ correct solution

③ correct angle $\angle BCD$ ② substantial attempt using cosine rule to find $\angle BCD$ ① finding length of DB 

Use the information provided to find the *actual* bearing on which she was walking from C to D , correct to the nearest degree.

$$\tan 28^\circ = \frac{48}{x}$$

$$x = \frac{48}{\tan 28^\circ}$$

$$= 90.2748 \dots$$

$$= 90.27 \text{ m (2 d.p.)}$$

$$\angle BCD : \cos(\angle BCD) = \frac{50^2 + 120^2 - (90.27)^2}{2 \times 50 \times 120}$$

$$\angle BCD = 43.1741 \dots^\circ$$

$$\text{Bearing from } C \text{ to } D = 270^\circ - 43.1741 \dots^\circ$$

$$= 226.8258 \dots^\circ \text{ (nearest degree)}$$

Question 40 (4 marks)

Solve the trigonometric equation $2 \sin^2 2x - \sin 2x = 1$ for $0 \leq x \leq 2\pi$.

4

$$2 \sin^2 2x - \sin 2x - 1 = 0$$

$$(2 \sin 2x + 1)(\sin 2x - 1) = 0$$

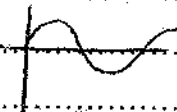
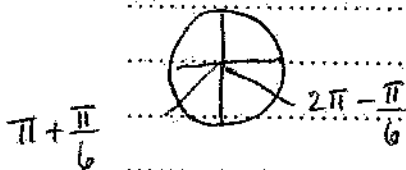
$$\sin 2x = -\frac{1}{2} \quad \text{or} \quad \sin 2x = 1$$

$$0 \leq x \leq 2\pi$$

$$0 \leq 2x \leq 4\pi$$

$$\sin^{-1}\left(-\frac{1}{2}\right) = \frac{7\pi}{6}$$

$$\sin^{-1}(1) = \frac{\pi}{2}$$



$$2x = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \frac{23\pi}{6} \quad \text{or} \quad 2x = \frac{\pi}{2}, \frac{5\pi}{2}$$

$$\therefore x = \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12} \quad \text{or} \quad x = \frac{\pi}{4}, \frac{5\pi}{4}$$

(4) correct sol.

(3) substantial solution, not all sol. found.

(2) correct domain/factored

(1) correct change of domain

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

1. The first part of the document discusses the importance of maintaining accurate records of all transactions and activities. It emphasizes the need for transparency and accountability in financial reporting.

2. The second part of the document outlines the various methods and techniques used to collect and analyze data. It includes a detailed description of the experimental procedures and the statistical analysis performed.

3. The third part of the document presents the results of the study. It includes a series of tables and graphs that illustrate the findings of the research.

4. The fourth part of the document discusses the implications of the findings and provides recommendations for future research. It also includes a conclusion that summarizes the main points of the study.

5. The fifth part of the document contains a list of references and a list of figures. The references include a list of books, articles, and other sources used in the study. The figures include a list of tables and graphs that are included in the document.

6. The sixth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

7. The seventh part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

8. The eighth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

9. The ninth part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

10. The tenth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

11. The eleventh part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

12. The twelfth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

13. The thirteenth part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

14. The fourteenth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

15. The fifteenth part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

16. The sixteenth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

17. The seventeenth part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

18. The eighteenth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

19. The nineteenth part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

20. The twentieth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

21. The twenty-first part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

22. The twenty-second part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

23. The twenty-third part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

24. The twenty-fourth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

25. The twenty-fifth part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

26. The twenty-sixth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

27. The twenty-seventh part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

28. The twenty-eighth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

29. The twenty-ninth part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

30. The thirtieth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

31. The thirty-first part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

32. The thirty-second part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

33. The thirty-third part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

34. The thirty-fourth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

35. The thirty-fifth part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

36. The thirty-sixth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

37. The thirty-seventh part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

38. The thirty-eighth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

39. The thirty-ninth part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

40. The fortieth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

41. The forty-first part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

42. The forty-second part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

43. The forty-third part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

44. The forty-fourth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

45. The forty-fifth part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

46. The forty-sixth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.

47. The forty-seventh part of the document contains a list of footnotes. The footnotes include a list of tables and graphs that are included in the document.

48. The forty-eighth part of the document contains a list of references. The references include a list of books, articles, and other sources used in the study.

49. The forty-ninth part of the document contains a list of figures. The figures include a list of tables and graphs that are included in the document.

50. The fiftieth part of the document contains a list of appendices. The appendices include a list of tables and graphs that are included in the document.